

Hallucination in Large Language Models and Retrieval-Augmented Generation: Mechanisms, Mitigation, and Evaluation

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Abstract. Large language models have demonstrated strong generative capability in question answering, dialogue, and other knowledge-intensive tasks. However, their outputs remain vulnerable to hallucination, including factual errors, unsupported claims, spurious citations, and distorted reasoning. Retrieval-augmented generation (RAG) has been proposed as a practical remedy because it supplements parametric knowledge with external evidence retrieved at inference time. Yet RAG does not guarantee truthfulness or attribution by default. Errors may arise during query formulation, document retrieval, evidence aggregation, and answer grounding. This paper reviews the relationship between hallucination and RAG from three perspectives: mechanism, mitigation, and evaluation. First, the major causes of hallucination in both vanilla LLMs and RAG-enhanced systems are analyzed. Second, the principal mitigation strategies are organized into retrieval optimization, evidence-grounded generation, and post-generation verification. Third, the main evaluation dimensions are examined, including factuality, faithfulness, attribution quality, and retrieval relevance. It is argued that RAG should not be treated as a complete solution to hallucination. Its value lies in enabling externally grounded generation, but its effectiveness depends on the reliability of retrieval, the fidelity of evidence use, and the rigor of evaluation. Future work should prioritize attribution-aware generation, conflict-sensitive reasoning, and unified evaluation protocols for trustworthy LLM systems.

Keywords: LLM, RAG, Hallucination

1. Introduction

Large language models have reshaped natural language processing by showing strong performance across open-domain question answering, dialogue, summarization, and tool-assisted reasoning. Nevertheless, this performance has been accompanied by a persistent failure mode: hallucination. In this context, hallucination refers to generated content that is fluent and plausible in form but unsupported, incorrect, or fabricated in substance. The problem is not restricted to isolated factual mistakes. It also includes missing evidence, non-faithful synthesis, incorrect source attribution, and reasoning trajectories that appear coherent while remaining unsound [1, 2]. The significance of

hallucination has increased as LLMs have been deployed in knowledge-intensive settings. In high-stakes domains such as medicine, law, finance, and scientific assistance, unsupported outputs are not minor imperfections; they constitute direct risks to reliability and accountability. The conventional cause is often described as overreliance on parametric memory. Since model knowledge is compressed into parameters during pretraining, it may be incomplete, outdated, or poorly calibrated for long-tail facts [3]. As a result, the model may answer confidently even when adequate support is absent.

Retrieval-augmented generation has therefore been introduced as an important architectural response. Instead of relying only on internal parameters, RAG retrieves external passages or documents at inference time and conditions generation on those materials [4]. The core intuition is direct: if the model is supplied with relevant evidence, unsupported fabrication should be reduced. This idea has motivated work on retrieval design, evidence fusion, self-reflection, and grounded answer generation [5-8]. However, the effectiveness of RAG has often been overstated. Although retrieval may reduce hallucination caused by missing knowledge, it also introduces new failure points. Query formulation may drift from user intent, retrieval may surface irrelevant or incomplete evidence, and generation may ignore or distort retrieved passages. Moreover, answers may be accompanied by citations that do not actually support the stated claims [9, 10]. Thus, the central question is not whether RAG helps in principle, but under what conditions it produces grounded, faithful, and attributable outputs. The present draft is revised from the uploaded project report and reorganized around an explicit pipeline logic. To make this logic explicit, the Retrieval-Augmented Generation function can be articulated as an equation: $y = G(x, \varepsilon(x))$ where x denotes the user query, $\varepsilon(x)$ denotes the evidence retrieved for x , and $G()$ denotes the generator. In a non-RAG setting, $\varepsilon(x)$ is absent and generation relies primarily on parametric memory. In a RAG setting, truthfulness depends not only on the model $G()$, but also on the quality of $\varepsilon(x)$ and the faithfulness with which it is used. Accordingly, this paper is organized around three dimensions: mechanisms of hallucination, RAG-based mitigation strategies, and evaluation criteria for grounded generation. Aims to establish a multi-dimensional evaluation framework encompassing authenticity, fidelity, attribution quality, and retrieval relevance, thereby providing a research framework and future directions for building more credible and reliable generative AI systems.

2. Mechanisms of hallucination in llms and RAG systems

2.1. Hallucination in parametric generation

Hallucination in standard LLMs is rooted in the mismatch between probabilistic next-token prediction and factual correctness. The model is optimized to continue text plausibly, not to verify claims against an external world model. Consequently, when internal knowledge is weak, approximate, or stale, fluent error may be generated rather than explicit uncertainty [1, 2]. This can be expressed by the standard decoding objective. The process is shown in Equation (1).

$$\hat{y} = \operatorname{argmax} P_{\theta}(y|x) \quad (1)$$

where θ denotes model parameters. Equation (1) reveals the core limitation: the model maximizes likelihood under learned distributions, not factual support under external evidence.

2.2. Pipeline-level hallucination in RAG

Within RAG systems, the problem is redistributed across a multi-stage pipeline rather than removed. Four failure sources are especially relevant. First, pre-retrieval distortion may occur during query construction. User intent can be misread, conversation history can be compressed inaccurately, and reformulated queries may emphasize the wrong entities or constraints. The retrieval stage is then misdirected from the beginning. Second, retrieval-side hallucination arises when relevant evidence is not retrieved or noisy evidence is retrieved instead. Sparse retrieval may miss semantically related passages, whereas dense retrieval may return passages that are semantically similar but factually inappropriate. Chunking granularity also affects support quality [5, 11]. Third, evidence-integration failure occurs when retrieved material is not used faithfully. Evidence may be ignored, partially used, over-generalized, or merged with parametric memory in a way that changes meaning. In multi-document settings, conflicts between sources may be resolved incorrectly or not resolved at all [7–9]. Fourth, attribution hallucination occurs after generation. The answer may include citations or snippets that only partially support the claims, or fail to support them entirely. Citation presence therefore should not be equated with evidential sufficiency [10, 12]. To formalize this pipeline, let the retriever be denoted by R and the $topk$ evidence set by Equation (2):

$$\varepsilon_k(x) = TopK_k(\{d \in D | s(x, d)\}) \quad (2)$$

where $s(x, d)$ is a relevance score over corpus D . Hallucination risk is then not solely a function of generation, but of the coupled system. The process is shown in Equation (3):

$$H = f(R(x), G(x, \varepsilon(x)), A(y, \varepsilon(x))) \quad (3)$$

where $A()$ denotes attribution quality. Equation (3) clarifies that hallucination in RAG is compositional.

2.3. A structured taxonomy

For a short review, the error space may be condensed into the taxonomy shown in Table 1. This structure is preferable to a flat list because it aligns mechanisms, symptoms, and likely intervention points.

Table 1. Compact taxonomy of hallucination in RAG-enhanced LLMs

Stage	Typical failure	Consequence
Pre-retrieval	Query drift, intent loss, poor reformulation	Wrong evidence is requested
Retrieval	Low recall, noisy top- k , bad chunking	Relevant support is absent or diluted
Grounding	Evidence ignored, conflict unresolved, over-generalization	Output deviates from the retrieved support
Attribution	Unsupported citation, partial citation, fabricated support	Response appears trustworthy without being supported

3. RAG-based mitigation strategies

3.1. Multistage strategy to improve recall and precision

The first mitigation category is retrieval optimization. Early work established the importance of dense retrieval for open-domain question answering [11]. Subsequent systems adopted hybrid retrieval, query expansion, and reranking to improve evidence recall and precision. The objective has been to increase the probability that generation is conditioned on relevant passages rather than distractors. More recent work has explored iterative retrieval, where generation and retrieval interact over multiple rounds instead of a single retrieve-then-generate pass [7, 8]. The useful abstraction is the retrieval utility. The process is shown in Equation (4):

$$U_{ret} = aRecall@k + \beta Precision@k, a + \beta = 1, a, \beta \in [0, 1] \quad (4)$$

where a and β weight completeness and purity of evidence. In hallucination mitigation, high recall alone is insufficient if noisy passages dominate the context window.

3.2. Decoding based on evidence fidelity and self-reflection

The second category is evidence-grounded generation. The issue here is not retrieval alone, but whether the model remains faithful to retrieved evidence. Architectures such as Fusion-in-Decoder improved multi-passage integration [6]. REPLUG showed that retrieval can improve grounding even when the generator itself is not retrained [13]. Self-RAG further introduced self-reflection signals to determine when retrieval is necessary, whether evidence is sufficient, and whether the response should be revised [8]. This process can be described as minimizing a joint loss. The process is shown in Equation (5):

$$L = L_{gen} + \lambda_1 L_{faith} + \lambda_2 L_{attr} \quad (5)$$

where $\mathcal{L}_{gen}$ measures generation quality, $\mathcal{L}_{faith}$ penalizes deviation from evidence, and $\mathcal{L}_{attr}$ penalizes unsupported attribution. Equation (5) is not tied to a single model; it captures the design principle required by trustworthy RAG.

3.3. Trust assessment mechanism to ensure the reliability of answers

The third category is verification and refusal. Since neither retrieval nor grounded decoding fully prevents unsupported generation, an additional verification layer is increasingly adopted. SelfCheckGPT showed that hallucination can be detected by examining consistency across alternative generations [14]. FactScore introduced fine-grained factual evaluation based on atomic claims [15]. In RAG systems, this logic supports answer-evidence consistency checking, citation verification, and abstention when evidence is insufficient. A post-generation trust score can be expressed as Equation (6) below:

$$T(y) = \gamma_1 F(y) + \gamma_2 S(y, \varepsilon_k) + \gamma_3 C(y, \varepsilon_k) \quad (6)$$

where $F(y)$ denotes factuality, $S(y, \varepsilon_k)$ denotes support or faithfulness to retrieved evidence, and $C(y, \varepsilon_k)$ denotes citation quality. In practice, $\gamma_i \geq 0$ and $\sum \gamma_i = 1$. When $T(y)$ falls below a threshold τ , the system should abstain from Equation (7):

$$output = \begin{cases} y & T(y) \geq \tau, \\ \text{"insufficient evidence"} & T(y) < \tau \end{cases} \quad (7)$$

The practical implication is straightforward. Retrieval optimization without grounded decoding improves the quality of ignored evidence. Grounded decoding without reliable retrieval amplifies bad evidence. Verification without attribution cannot distinguish between plausible and genuinely supported outputs. Effective mitigation is therefore pipeline-level and compositional.

4. Evaluation of hallucination mitigation

Evaluation remains one of the weakest components of current RAG research. Traditional metrics such as exact match, F1, or answer accuracy are insufficient. A response may match a gold answer while relying on irrelevant evidence; conversely, a partially correct answer may be more trustworthy if it is transparently grounded. The first major dimension is factuality, namely whether the generated claims are true. Recent work such as FactScore, evaluates long-form outputs at the level of atomic statements [15]. The second dimension is faithfulness, namely whether the answer is supported by the provided evidence. An answer may be factually correct in the world yet still not be grounded in the retrieved passages [9, 16]. The third dimension is attribution quality, which includes citation precision, citation coverage, and evidence sufficiency [10]. The fourth dimension is *retrieval quality*, since evidence cannot support generation if it was never retrieved [5].

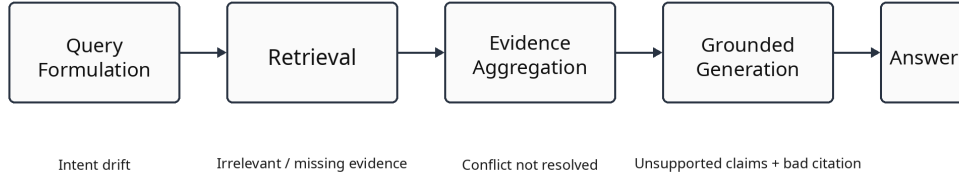


Figure 1. Pipeline view of hallucination in retrieval-augmented generation.(picture credit: original)

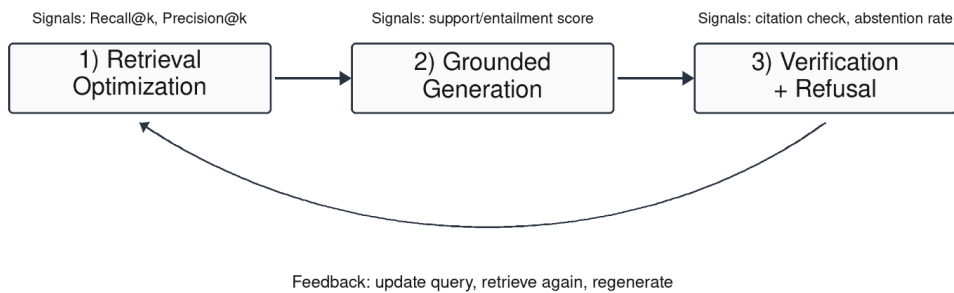


Figure 2. Closed-Loop mitigation mechanism for retrieval-augmented generation.(picture credit: original)

A compact multi-objective evaluation score may therefore be written as Equation (8):

$$Q_{RAG} = w_1 Q_{fact} + w_2 Q_{faith} + w_3 Q_{attr} + w_4 Q_{ret} \quad (8)$$

Where Q_{fact} , Q_{faith} , Q_{attr} , and Q_{ret} denote factuality, faithfulness, attribution, and retrieval relevance, respectively, and $\sum iW_i = 1$. Equation (8) emphasizes that no single metric is adequate for trustworthy RAG assessment. To facilitate researchers in selecting appropriate evaluation tools in practice, Table 2 systematically organizes the above four core dimensions and their corresponding operational indicators, and supplements them with the indispensable human evaluation dimensions in large model evaluation.

Table 2. Key dimensions and indicators for hallucination evaluation in RAG

Dimension	Question	Typical indicator
Factuality	Is the claim true?	FactScore, claim verification
Faithfulness	Is the answer supported by retrieved evidence?	entailment/support score
Attribution	Do citations support the exact claim?	citation precision, coverage
Retrieval quality	Was the right evidence retrieved?	Recall@k, MRR, relevance
Human judgment	Is the response reliable in context?	expert annotation, pairwise preference

Automated metrics remain necessary but insufficient. Human evaluation is still required for subtle reasoning distortions, partial support, and misleading citation placement, especially in domain-specific applications [1, 16]. In this sense, Figure 1 provides the failure decomposition that motivates these metrics. Unreliable RAG assessments are inherently multi-objective because authenticity, support, citation accuracy, and retrieval adequacy reflect different failure modes. Figure 2 provides the operational loop to be measured.

5. Conclusion

Hallucination should be regarded as a structural problem of large language models rather than an incidental defect. Retrieval-augmented generation introduces an important corrective by making external evidence available at inference time. However, RAG does not remove hallucination by default. Instead, the locus of failure shifts from internal memory alone to the full generation pipeline, including query formation, retrieval, evidence integration, and attribution.

Based on the foregoing, the following conclusions can be drawn. First, mitigation should be treated as a pipeline problem rather than a retrieval problem alone. Second, grounded generation requires both relevant evidence and faithful use of that evidence. Third, evaluation should move beyond answer accuracy toward factuality, faithfulness, attribution, and retrieval relevance. RAG remains one of the most promising directions for reducing hallucination, but its success depends on whether trustworthy evidence can be retrieved, preserved, and audited throughout generation. Future work should therefore focus on attribution-aware architectures, conflict-sensitive evidence reasoning, and evaluation protocols that measure support rather than surface plausibility.

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